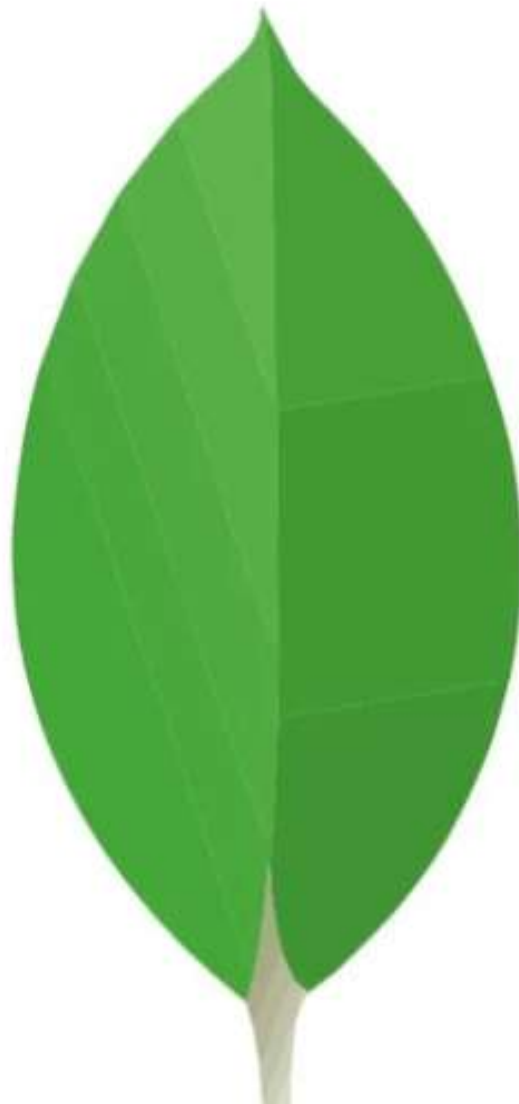


MONGODB

BEGINNER TO ADVANCED



DAY 1: INTRODUCTION TO MONGODB

What is MongoDB: Introduction to NoSQL databases, and how MongoDB differs from relational databases.

Document-Oriented: Understand MongoDB as a document-based database using JSON-like data.

Install MongoDB: Set up MongoDB on your machine or use MongoDB Atlas for a cloud-based solution.

Basic Commands: Start the MongoDB shell and learn basic commands like **show dbs**, **use**, and **show collections**.

DAY 2: INSERTING AND QUERYING DOCUMENTS

Documents: MongoDB stores data in BSON format (Binary JSON), and documents are the core data structure.

Insert Data: Learn how to insert single and multiple documents using **insertOne()** and **insertMany()**.

Query Documents: Retrieve documents from a collection using **find()**.

Comparison Operators: Use basic comparison operators like **\$gt**, **\$lt**, **\$eq**, **\$ne**.

DAY 3: UPDATING AND DELETING DOCUMENTS

Update Documents: Modify data using `updateOne()`, `updateMany()`, and `$set`.

Delete Documents: Remove documents using `deleteOne()` and `deleteMany()`.

Replace Documents: Use `replaceOne()` to replace entire documents.

Update Operators: Learn operators like `$inc`, `$min`, `$max`, and `$rename`.

DAY 4: QUERYING WITH CONDITIONS AND LOGICAL OPERATORS

Advanced Queries: Learn how to use multiple conditions in queries.

Logical Operators: Use logical operators like **\$and**, **\$or**, **\$not**, **\$nor** for complex queries.

Element Queries: Query fields based on their presence (**\$exists**) or data type (**\$type**).

Example: Retrieve students who have scored between 70 and 90 or have not submitted a particular assignment.

DAY 5: INDEXING FOR PERFORMANCE

Indexes: Understand the purpose of indexes and how they improve query performance.

Create Indexes: Use `createIndex()` to add single-field and compound indexes.

Unique Indexes: Ensure uniqueness for certain fields using unique indexes.

Text Indexes: Learn how to create text indexes for searching string data.

Index Management: View and drop indexes using `getIndexes()` and `dropIndex()`.

DAY 6: AGGREGATION FRAMEWORK

Aggregation: Understand the MongoDB aggregation framework for advanced data processing.

Pipeline Stages: Learn about stages like **\$match**, **\$group**, **\$sort**, **\$project**, **\$limit**, and **\$skip**.

Aggregation Operators: Use operators like **\$sum**, **\$avg**, **\$push**, **\$first**, **\$last**, and **\$addToSet**.

\$lookup: Perform joins between collections using **\$lookup**.

DAY 7: RELATIONSHIPS AND EMBEDDING DOCUMENTS

Data Modeling: Understand how to model relationships in MongoDB using embedding or referencing.

Embedded Documents: Store related data inside a document (denormalization).

Referencing: Use references (normalization) to link data across collections.

Pros and Cons: Learn the trade-offs between embedding and referencing.
